

Performance Verification

Stopwatch-timed performance verification testing across 3 runs each on the iOS Simulator (XCode) and Android Emulator (Android Studio) revealed consistent scores across each area, with the Chatbot response (reliant on the Gemini API) leading to the highest level of unpredictability. Some issues with the map loading time on Android require further development, averaging 15.0s versus 4.0s on iOS, nearly 4x slower, with one run failing to complete entirely. Two contributing factors were identified: first, the Android emulator's location services required manual GPS mocking during development (a known emulator limitation, not present on real devices), Further research indicated that the Android Emulator is known to be slower across the board in performance testing

(<https://developer.android.com/studio/run/emulator-acceleration>)

iOS Simulator testing

Test Item	Run 1 (s)	Run 2 (s)	Run 3 (s)	Average (s)
Map loading time	4.1	4.3	3.6	4.0
Report load on map	1.7	1.7	2	1.8
Live translation	2.7	5.3	4.8	4.3
Chatbot response	9.3	12.5	10.4	10.7
Account login	1.5	1.4	1.7	1.5
Account registration	3.2	3.0	3.5	3.2
Account logout	1.4	1.8	2.0	1.7
Account deletion	2.4	1.6	2.0	2.0

Android Emulator testing

Test Item	Run 1 (s)	Run 2 (s)	Run 3 (s)	Average (s)
Map loading time	12.2	FAIL	17.8	15.0
Report load on map	3.5	FAIL	3.1	3.3
Live translation	6.5	7.9	6.3	6.9
Chatbot response	11.8	14.5	10.5	12.3
Account login	2.3	2.9	2.5	2.6
Account registration	3.1	3.5	3.3	3.3
Account logout	3.3	3.8	3.1	3.4
Account deletion	2.8	2.9	4.0	3.2

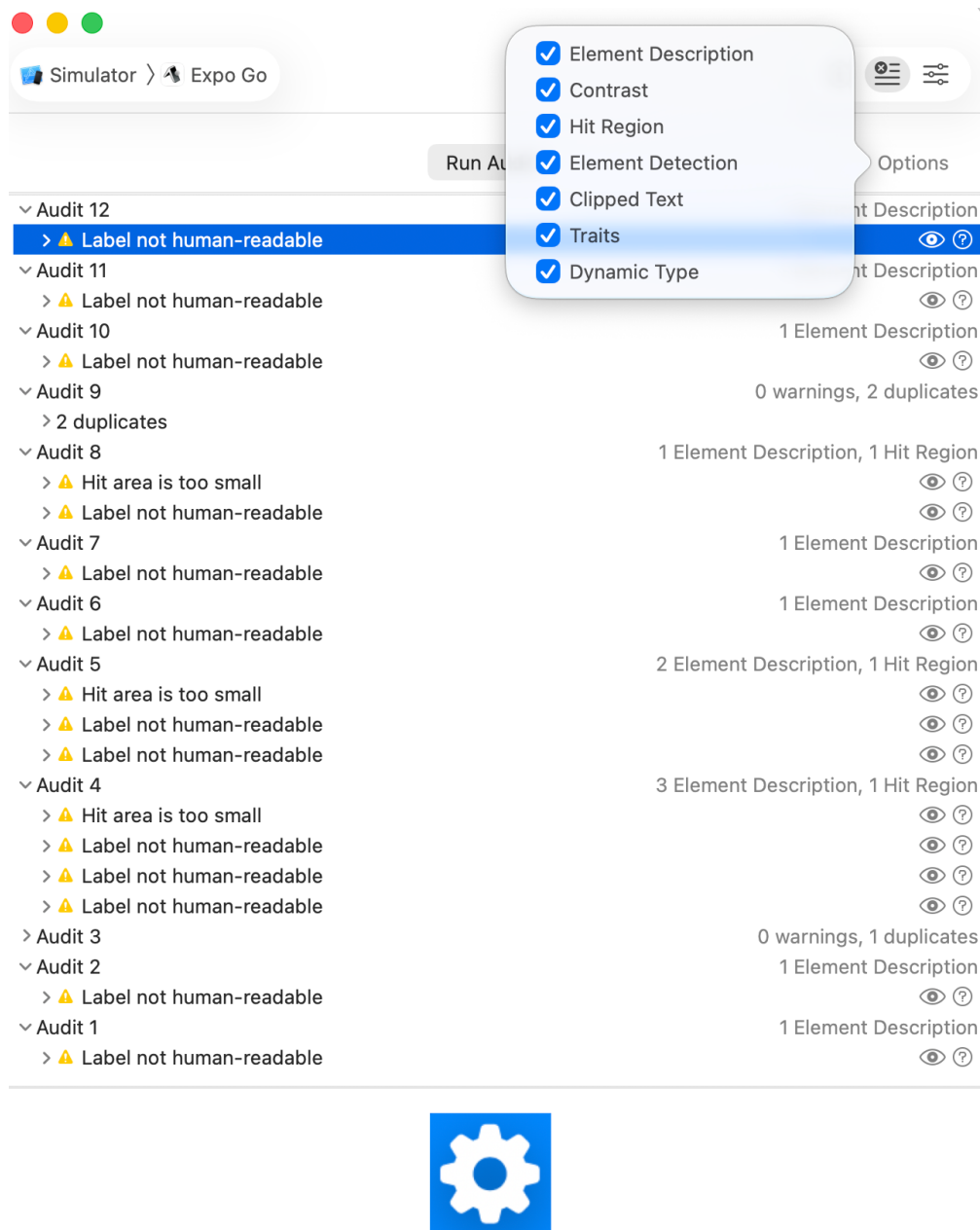
Accessibility Review

Accessibility Audit with XCode

Using XCode's Accessibility Inspector, it was found that there were some errors and labels that were not human-readable so that a Voice-Over application would not be able to understand. These point to any elements without readable text. 33

Accessibility Labels were added throughout the pages to facilitate this where buttons were not human-readable, and the sizes of certain elements were adjusted. However, as indicated in the audits below, the primary issue preventing a full pass on these tests actually came from the in-built ExpoGo tools button (the blue emoji in the screenshot overleaf), rather than from the ClearPath app itself.

In addition to labelling and hit-target issues, XCode's Accessibility Inspector audit also checks for colour contrast and clipped or truncated text. Across all audited screens, no contrast or text-clipping warnings were raised, indicating the app's existing colour palette and text layout already meet accessibility contrast minimums.



Text size

The ClearPath app supports dynamic text size as indicated in a selection of screenshots overleaf. In certain circumstances these cause small overlaps (dependent on language and the font size), but for the most part the system is prepared for a variety of font sizes across the 6 languages without blocking any buttons throughout the flow of the app.

